One year of contributions at Oslandia

Swiss QGIS User Group Meeting - 2026



OSLANDIA

Who are we?

- Oslandia
  - ~25 persons
  - Offer services in Open Source GIS
  - Contribute to Giro3D, QWC, SFCGAL, PostGIS, QGIS …
- 4 Speakers: QGIS core commiters


Jean


Loïc


Julien

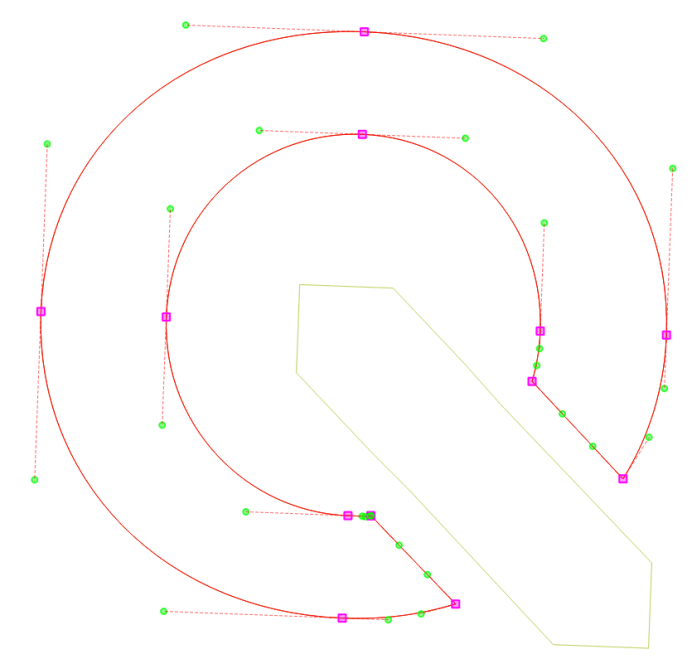

Benoît

Summary

Not all our contributions

- NURBS *Loïc Bartoletti*
- SFCGAL *Benoît De Mezzo*
- 3D styles *Jean Felder*
- Interface customization *Julien Cabieces*

NURBS (QGIS 4.0)

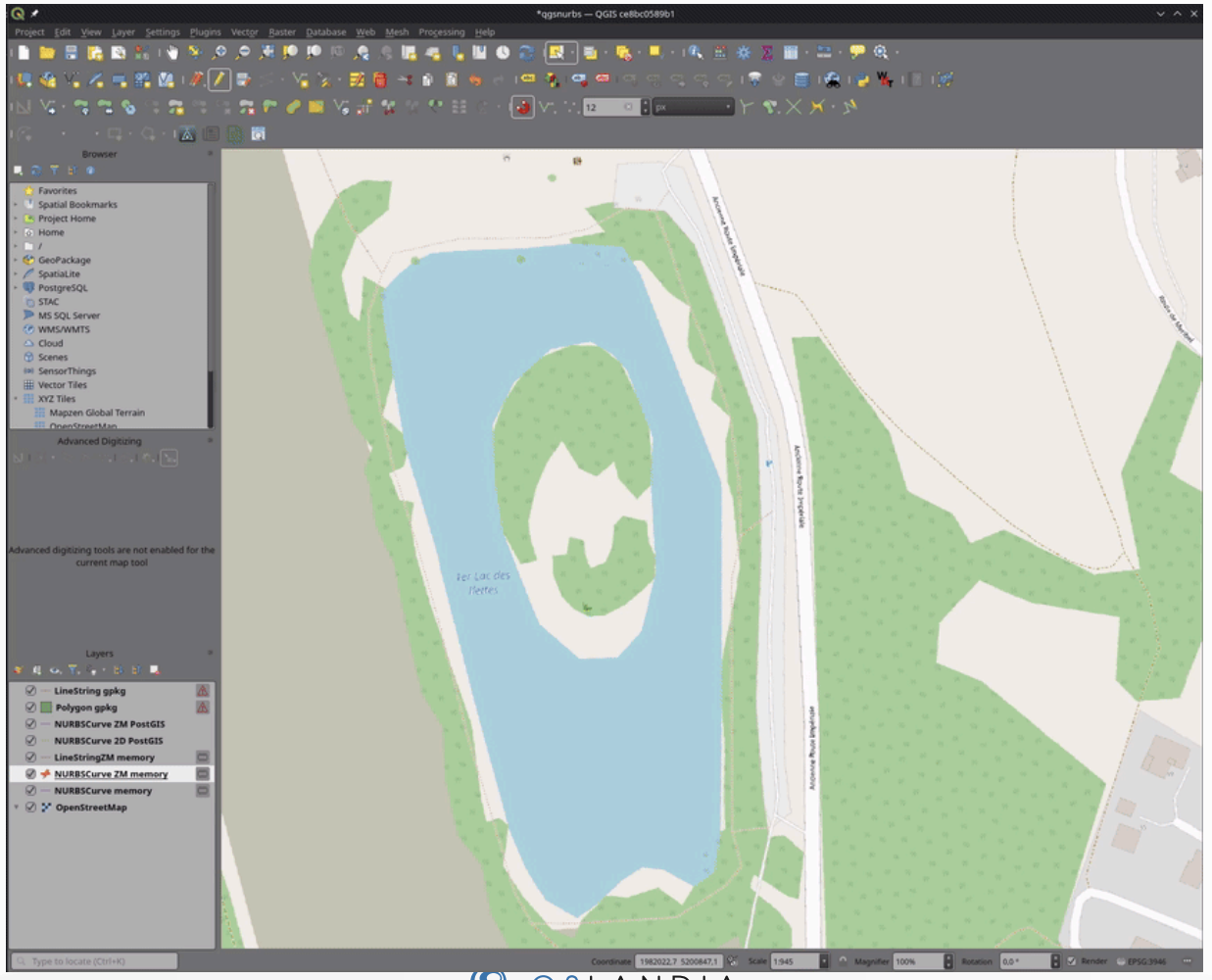


OSLANDIA

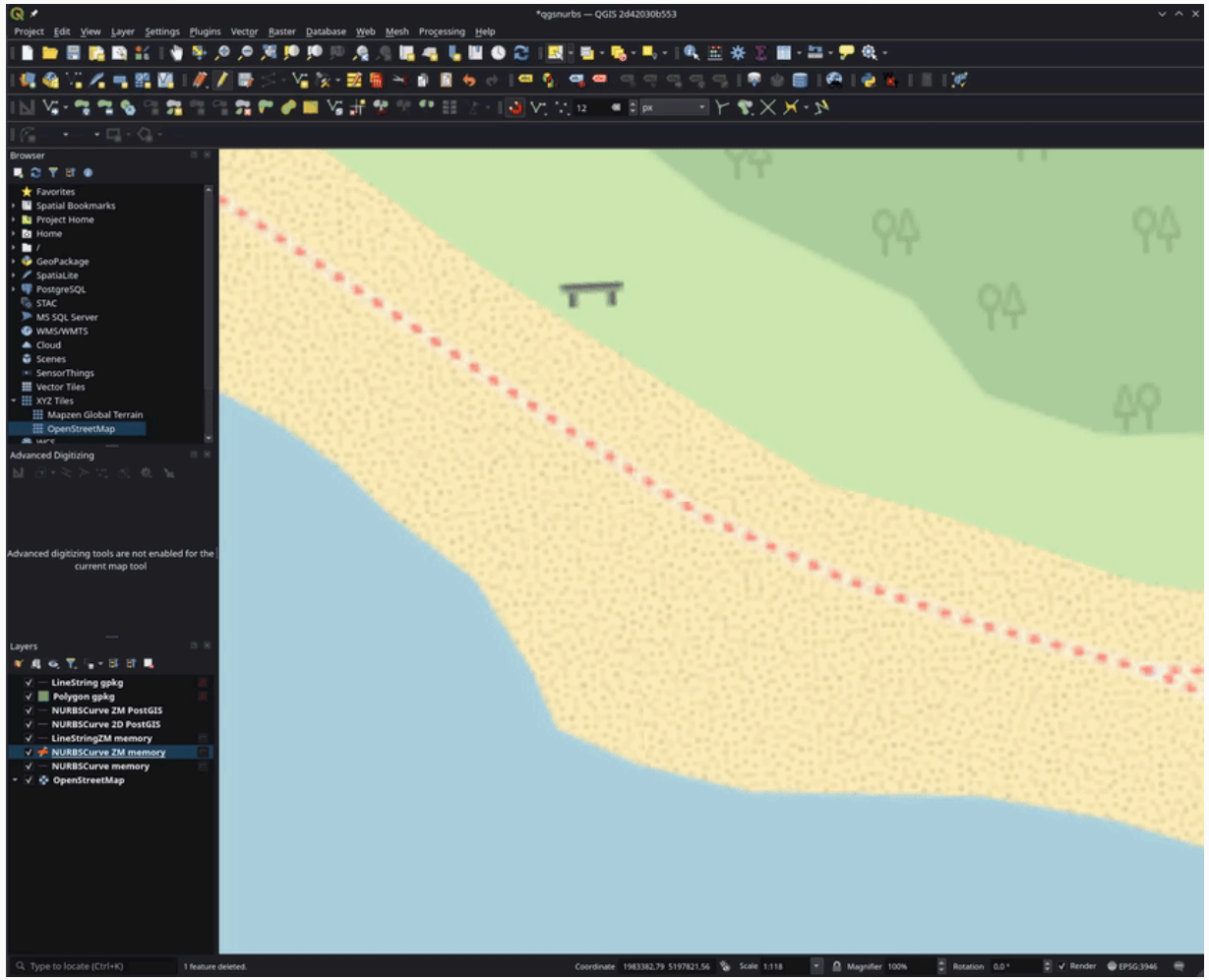
NURBS - What is it?

- Mathematical model to describe curves (and surfaces)
  - NURBS includes B-spline and Bézier curve
- Standard CAD and design tools
- and ISO (spatial) standard

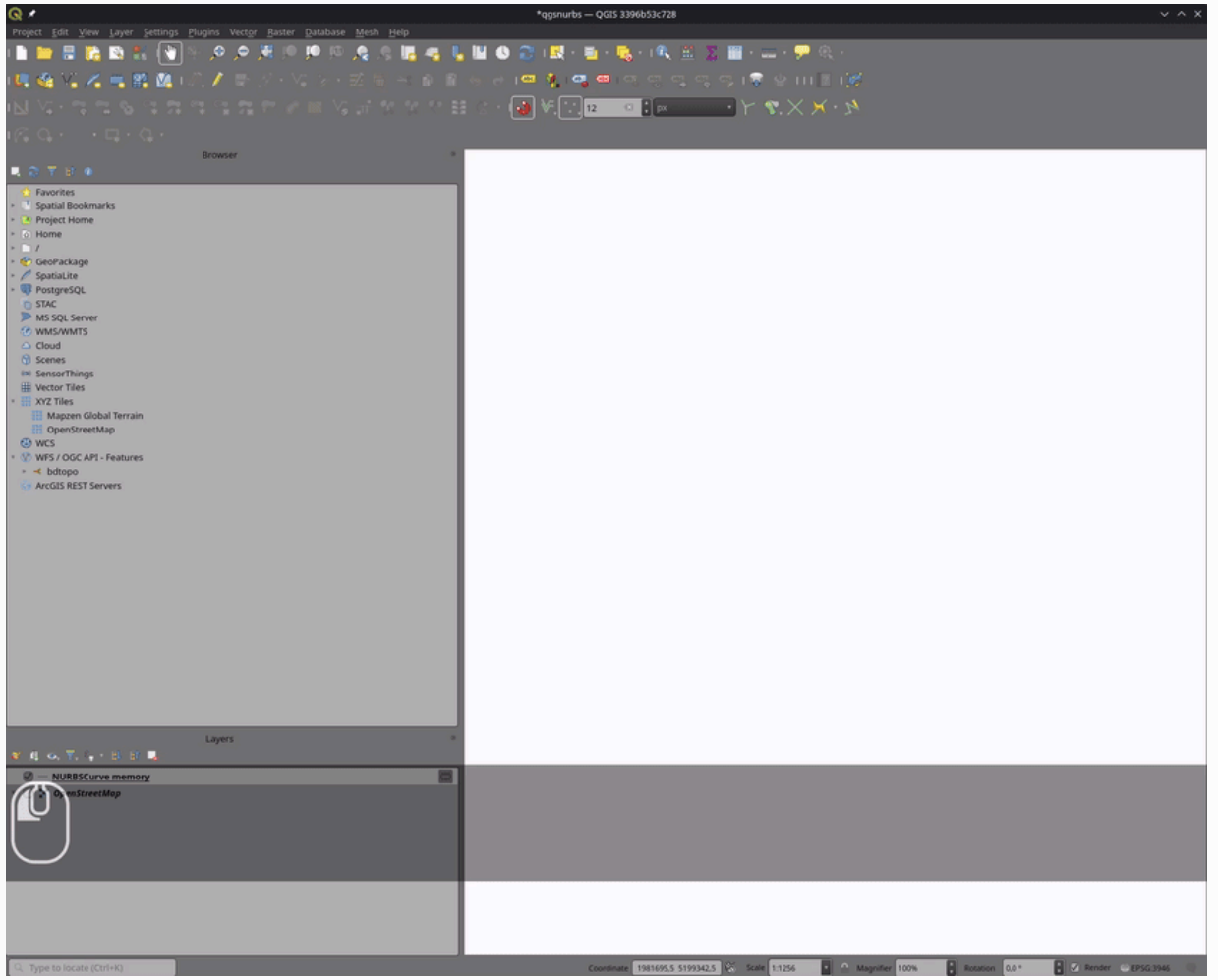
NURBS - Digitizing with handles/anchors (Bézier)



NURBS - Digitizing with control points



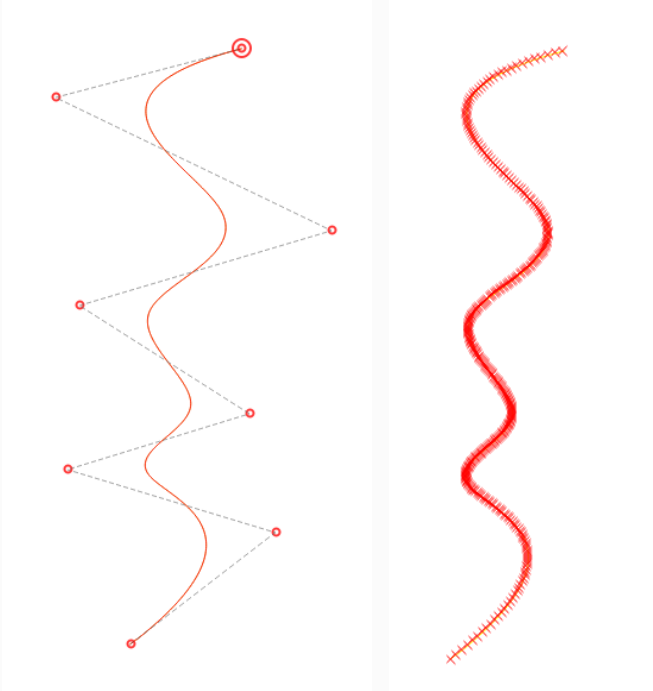
NURBS - Editing



NURBS - PostGIS Integration

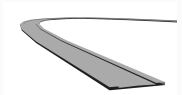
- QGIS : support in memory layer
- Other file format: segmentization
- PostGIS : native support in 3.7.0

NURBS - Comparison with/w.o support



NURBS - What's next

- Labelling
- Civil engineering



- 💸 Search for funding

SFCGAL (QGIS 4.2)



- Simple Feature over CGAL
- Computational Geometry Algorithms Library
  - 30th birthday
  - from academic researches
  - fully 3D and more
  - committed to precision

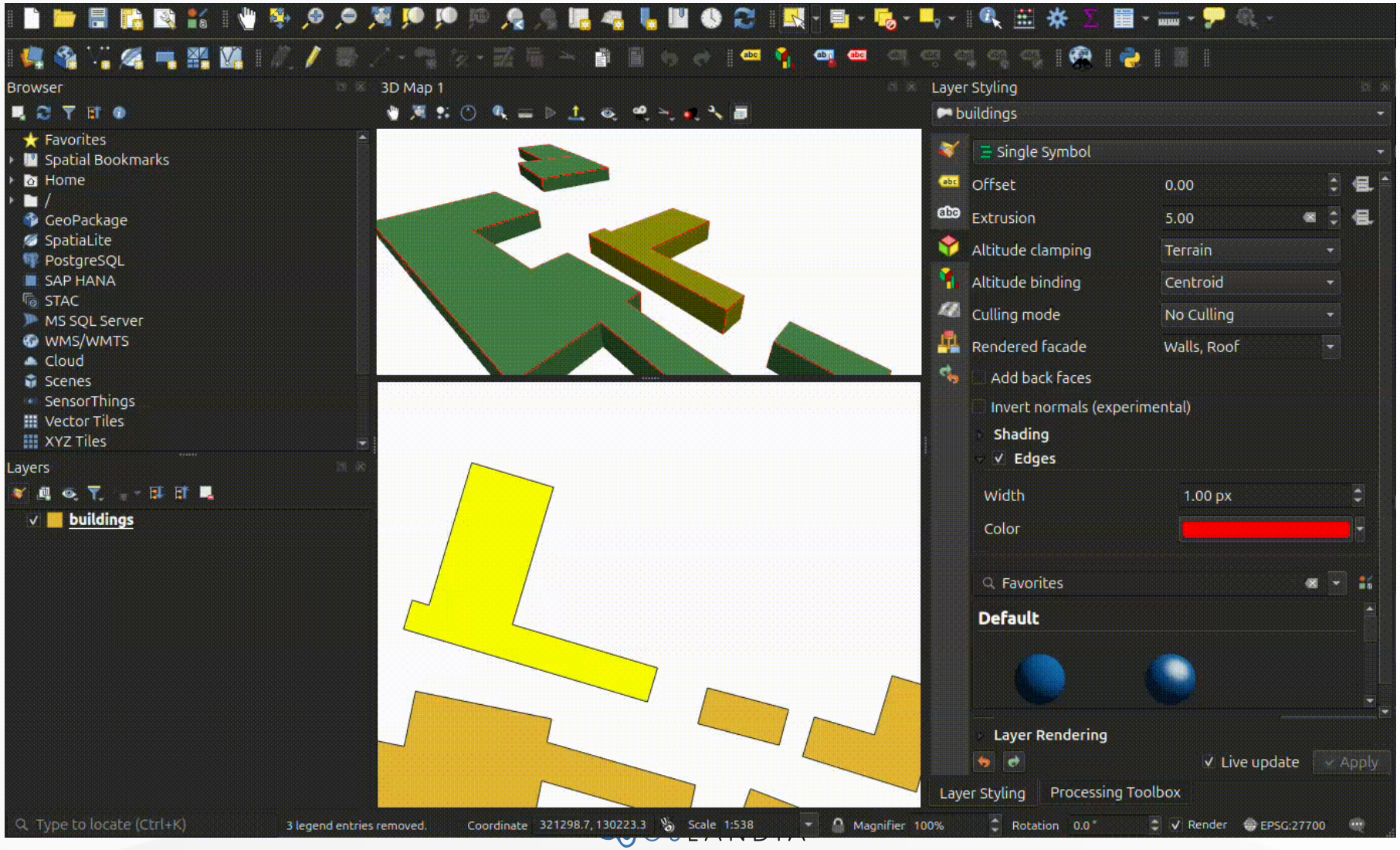
SFCGAL - Why in QGIS?

- GEOS
  - mainly in 2D
  - approximate or no support for 3D
- More and more full 3D usecases

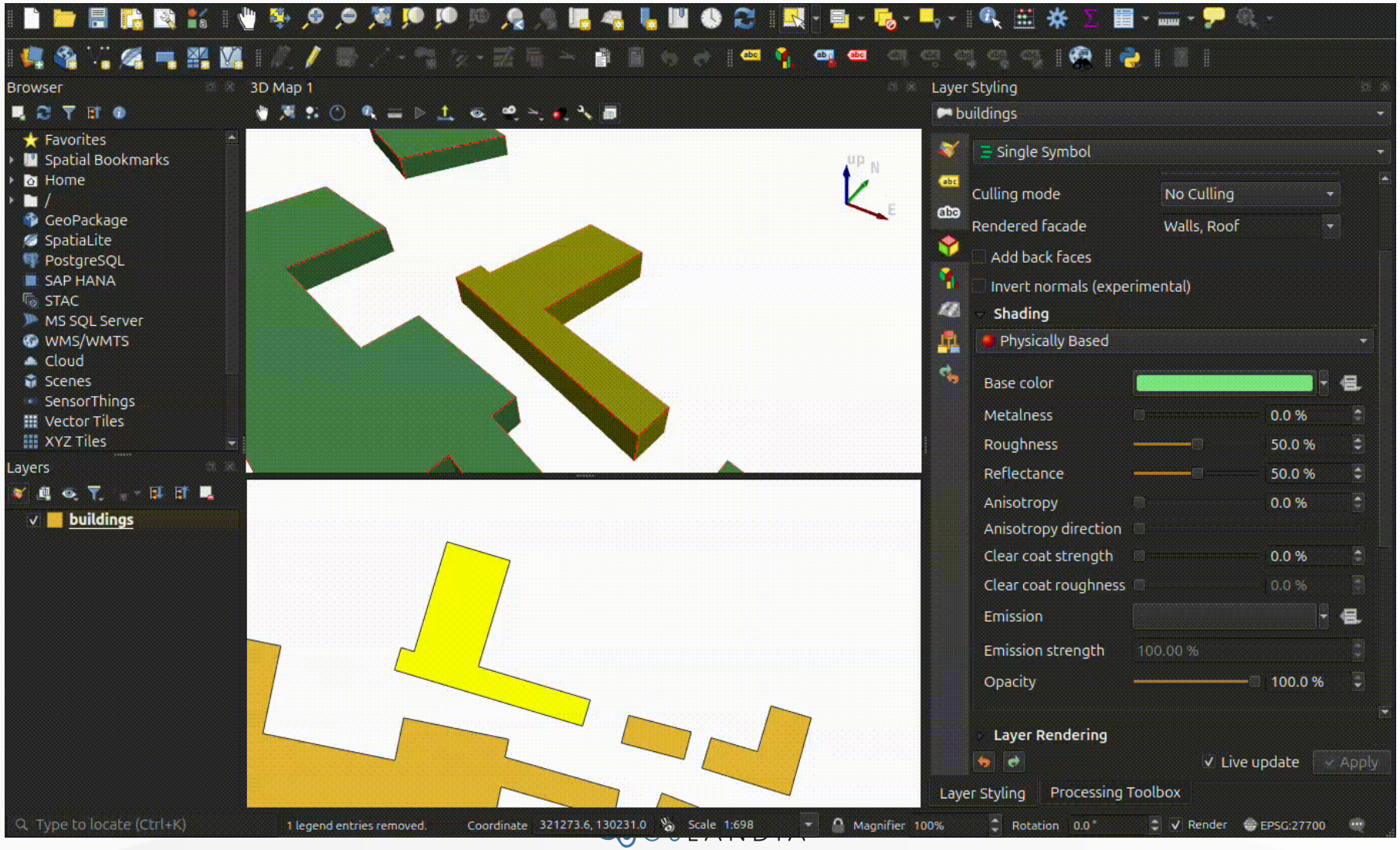
SFCGAL - Core integration

- v2.3.0 in QGIS 4.2 (python binded)
- Processings
  - Extrusion
  - Approximate medial axis
- Partial/Future integration
  - 3D editing toolbar
  - integrate QSFCGAL plugin into core
    - expressions
    - processings

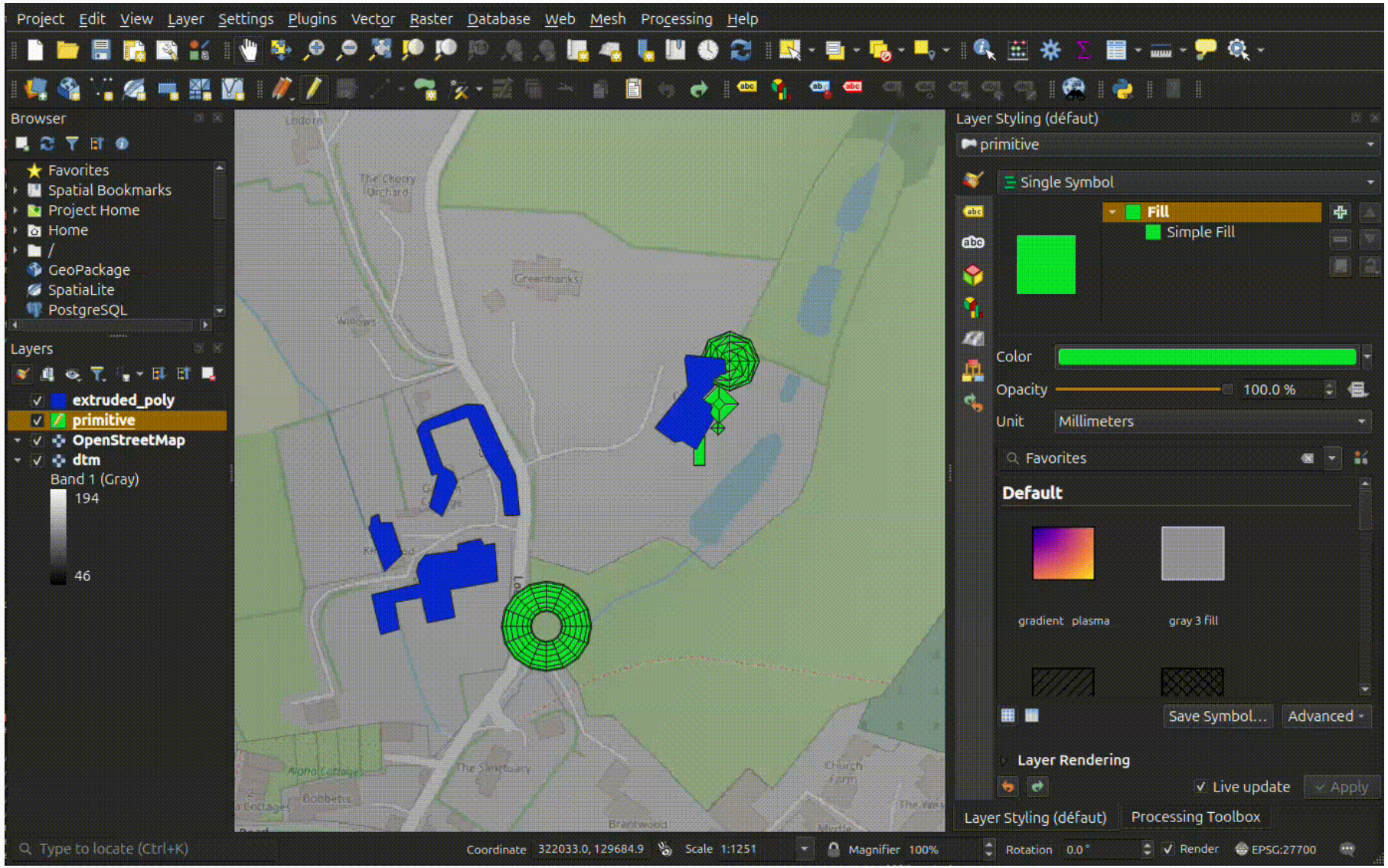
SFCGAL - Examples 1/3



SFCGAL - Examples 2/3

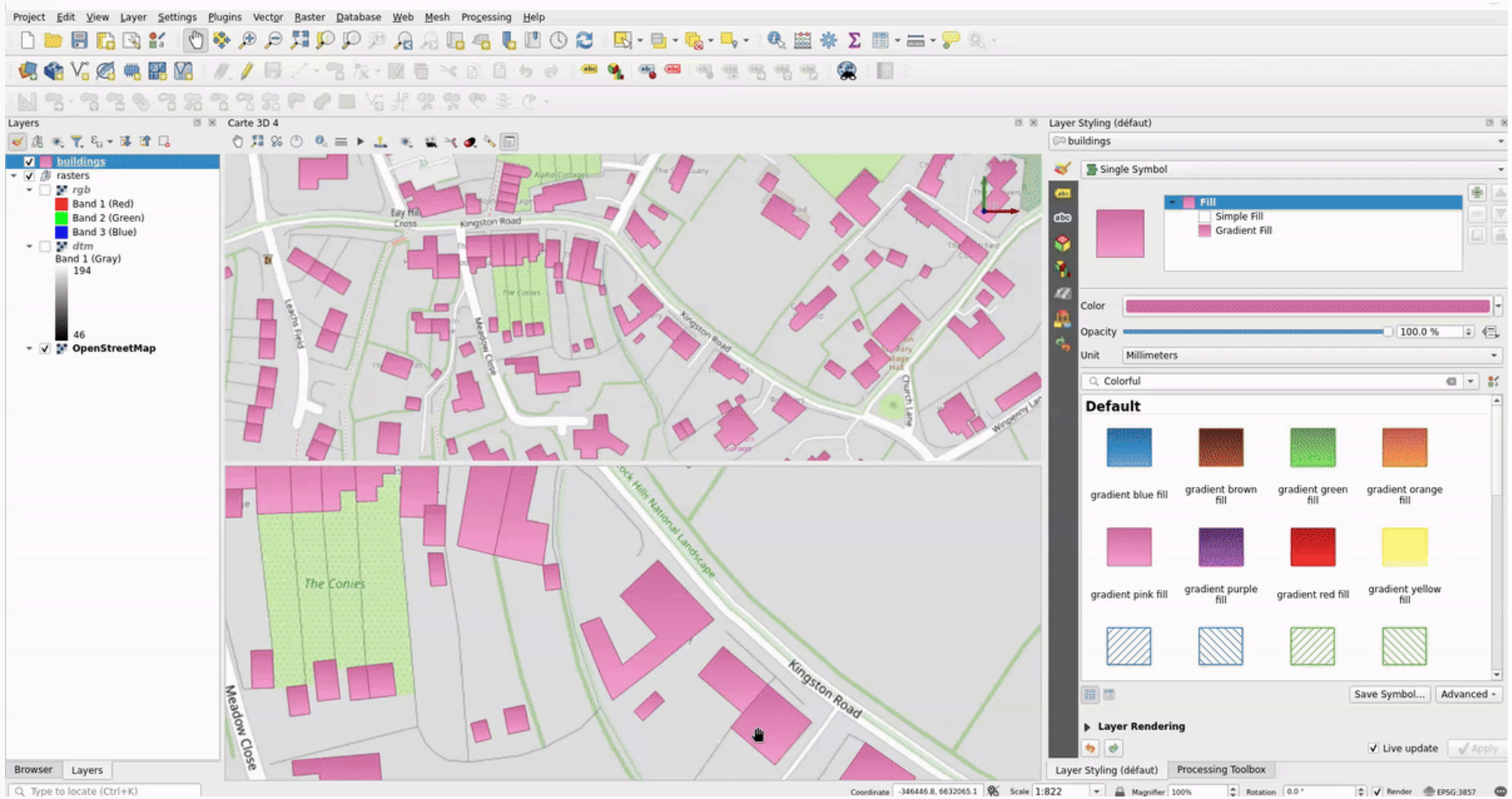


SFCGAL - Examples 3/3

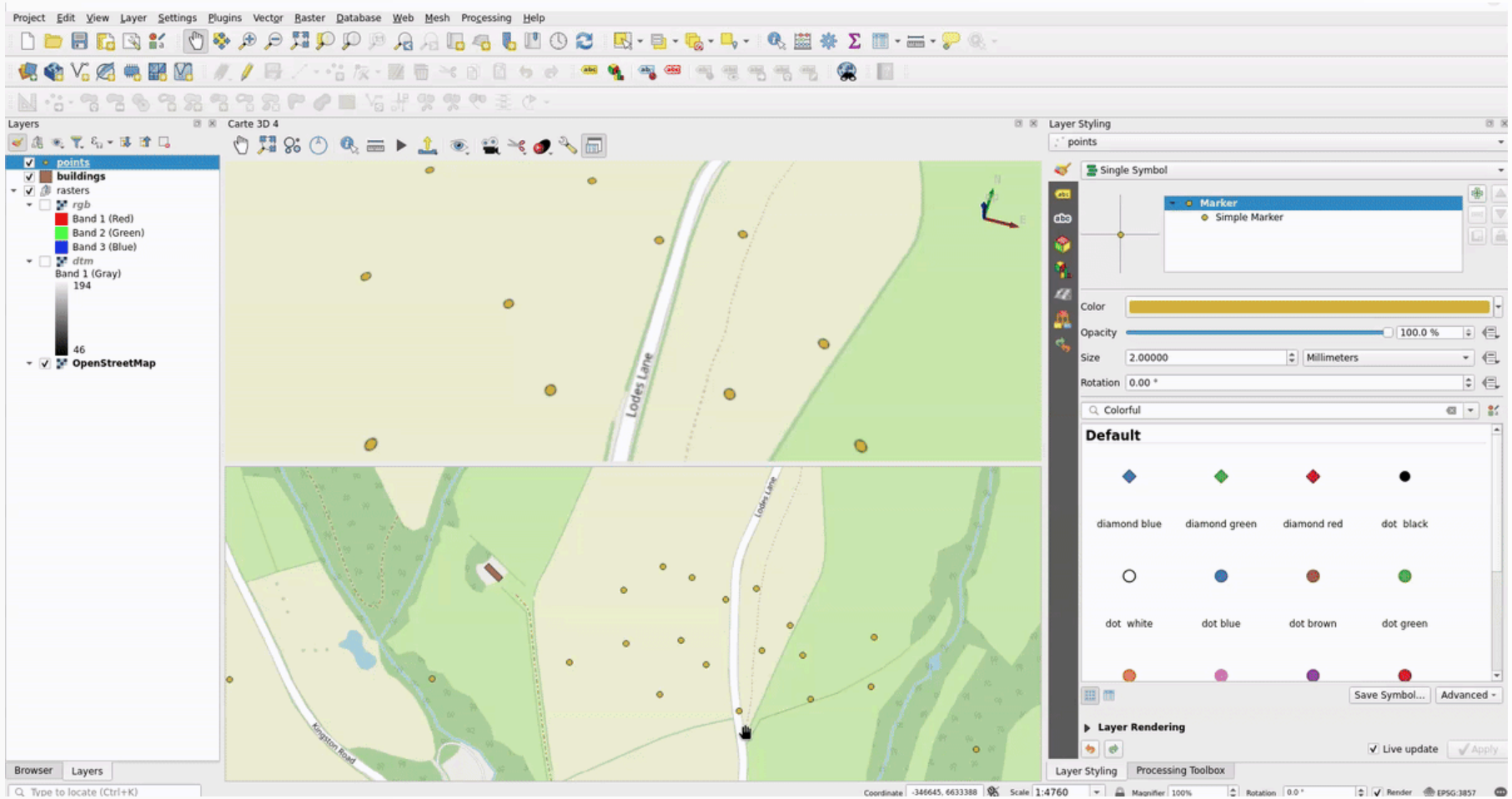


2D/3D Styles (QGIS 4.2)

3D Symbology by category

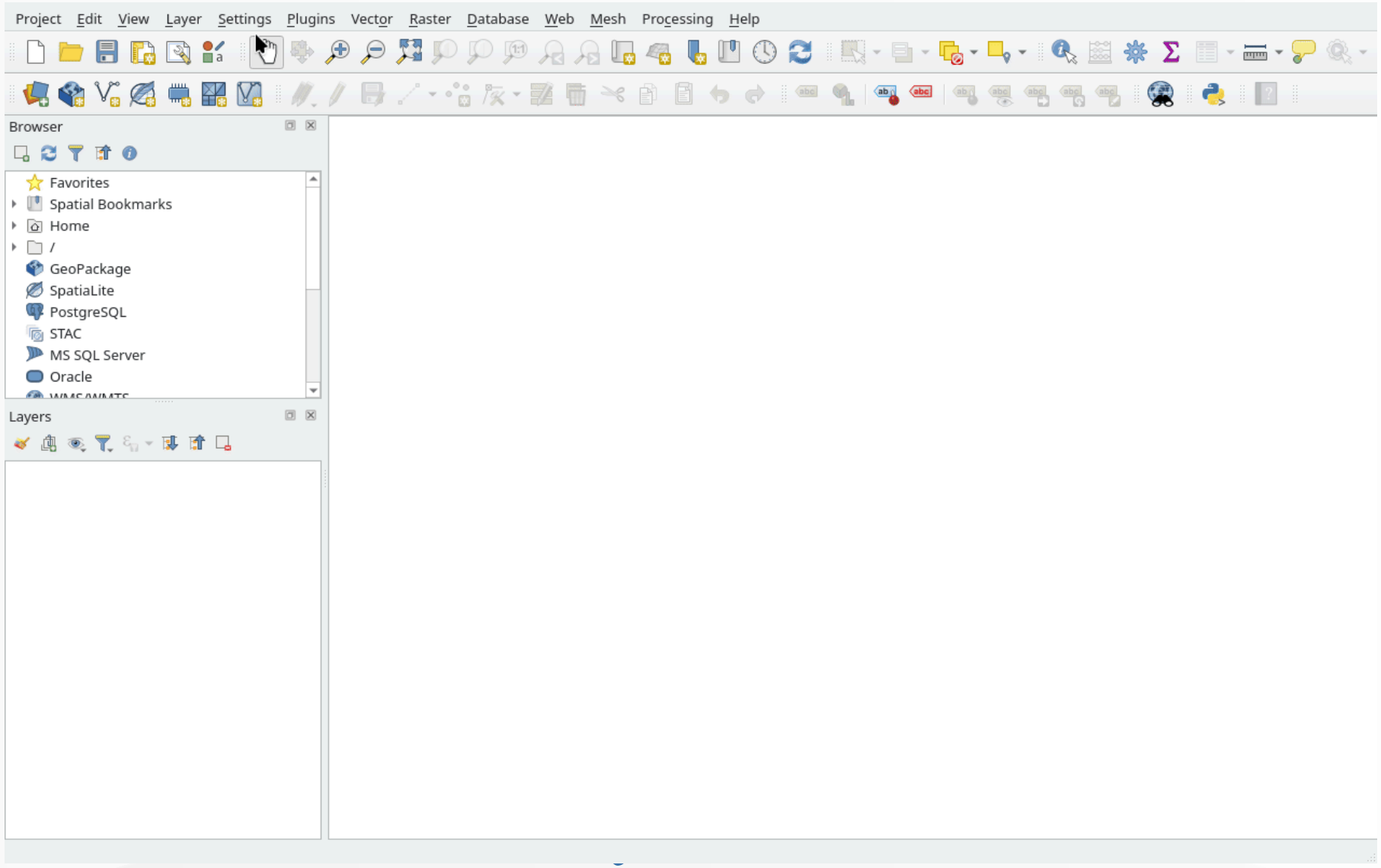


3D Symbology from 2D

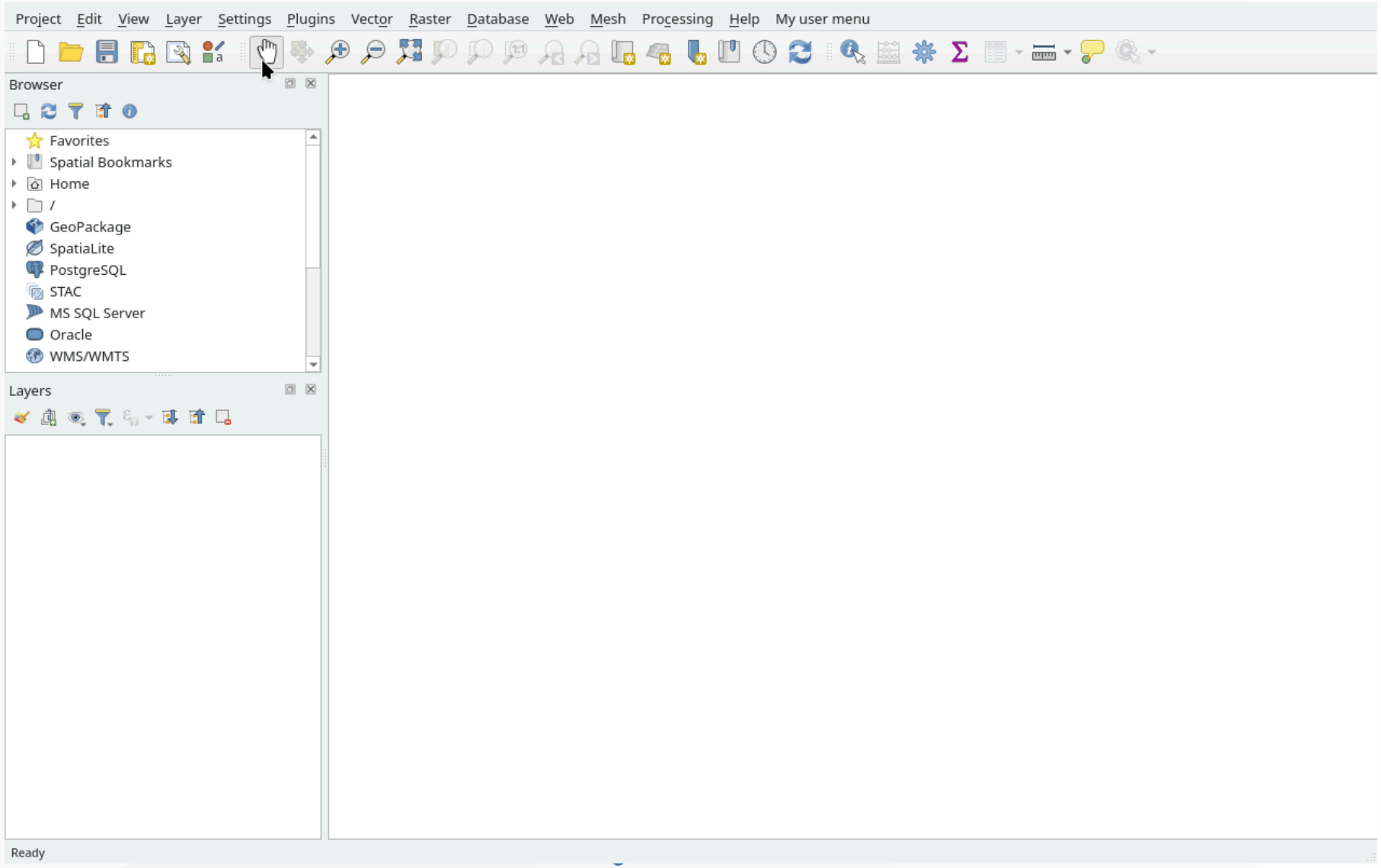


Interface customization (QGIS 4.2)

User defined menus



User defined toolbars



Interface customization

- ✅ Create/Delete your own menus/submenus and tool bars
- ✅ Complete rework of the code
- To prepare for “workspaces”
  - Presets of different interface customization
  - Switch between presets witout changing profile
- 💸 Search for funding

Thank you!

Want to know more? reach out at infos@oslandia.com